Internal Assessment: Mathematics, Analysis and Approaches

Introducing Inversion, Reflection Over a Circle.

Topic: Geometry, Complex Numbers

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1 Introduction

In Euclidean geometry, many statements become easy once the "right" viewpoint is chosen. A familiar example is reflection in a line: it preserves distances and angles, and it sends lines to lines and circles to circles. This makes it a reliable tool for turning a geometric condition into something you can directly read off from a diagram.

Inversion plays a similar role for circle geometry. It is a transformation defined relative to a chosen circle, and its key feature is that it maps every line or circle to another line or circle (with the special case that lines may become circles through the centre of inversion, and circles through the centre may become lines). Because of this, inversion is particularly effective when a configuration contains many circles and lines passing through the same point: it can convert a crowded picture into a cleaner one involving simpler objects relations.

This exploration introduces inversion from first principles, justifying its defining properties and proving its basic mapping rules algebraically on the complex plane. The goal is not to catalogue tricks, but to show why inversion is a natural "reflection across a circle" and why it deserves a place alongside more familiar transformations.

2 Inversion

Out of nothing I have created a strange new universe. János Bolyai

Many configurations in Euclidean geometry involve circles and lines that look complicated in the original picture but become simple after a suitable transformation. Inversion is one such transformation: it "reflects" points across a circle in a way that turns circles and lines into circles and lines, often converting difficult geometric conditions into more tractable incidence statements (or vice versa).

2.1 Motivation for Inversion

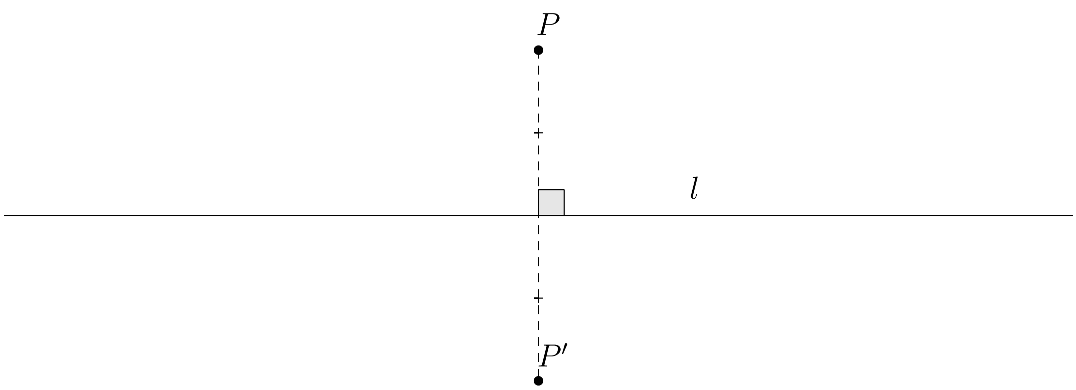

Figure 1: Reflecting a point over a line

Firstly, let us intuitively motivate the upcoming definition of an inversion. We can easily reflect a point across a given line: draw a normal from point onto the line, and mark the refection as the point on this normal which has an equal distance to the line. In **Figure** 1 point P' is the reflection point P across line l.

Now, what if instead of a line we had a circle instead? How should reflection be defined in that case? This premise may seem odd, but it will turn out to be a surprisingly powerful tool if defined appropriately.

Let us first try to define reflection over a circle in ‘the obvious way’: Like with the line, draw a normal from the point onto the circle, and mark the reflection as the point on this normal which has an equal distance to the point where the normal intersects the circle. This is quite simple to do, since the normal to the circle from a point is simply the line connecting it with the centre.

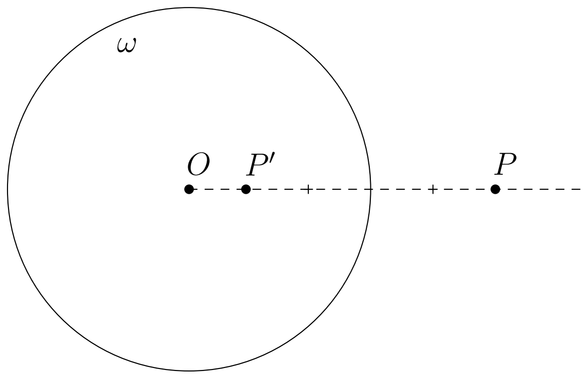

Figure 2: Naïvely reflecting over a circle

In **Figure** 2, point O is the centre of circle ω, and line OP is the normal drawn from point P. With this definition, point P' is the reflection of P across ω. Though this looks reasonable so far, upon attempting to make reflections for more points, we realise the drawbacks of such a guileless definition.

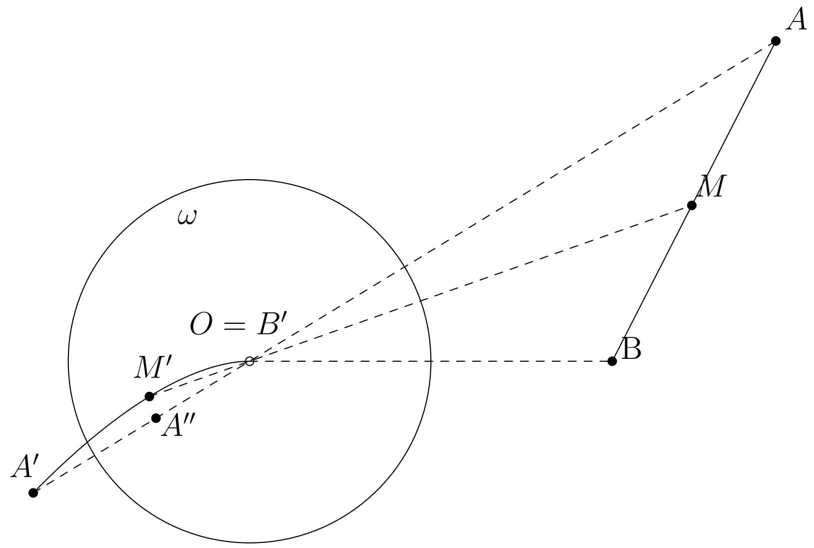

Figure 3: Exposing the weakness of the naïve definition

Again, points A', B' are the reflections of A, B, and likewise A'', B'' are the reflections of A', B'. But making observations from **Figure** 3 we realise some undesireable implications of this definition.

It feels wrong that A and A' lie on the same side of the circle (the two sides of a circle are 'inside' and 'outside') even though A' is the reflection of A. Moreover, though points A' and B' are the reflections of A and B, points A and B themselves are not the reflections of A' and B'. What even is the reflection of O? It is not possible to define it uniquely since it could be point B, or that point rotated with any angle. Worse still, if point M moves along the segment AB, then its reflection traces some complicated curve with endpoints A', B', instead of something familiar like a line or a circle.

This last issue in particular cripples this definition of reflecting over a circle. When we reflect many points over a line, they still stay the same shape: lines map to lines, circles map to circles; it is not possible for simple things to have complicated reflections.

We need to define the reflection over a circle in a more deliberate way to avoid these undesirable weaknesses. Since we call the operation 'reflection', it should share some of the most fundamental properties fulfilled when reflecting over a line l.

Property 1. The reflection of each point exists and is uniquely defined.

Property 2. Reflecting the reflection of a point yields the point itself. We call a map with this property an 'involution'.

Property 3. Points on the line l are fixed, i.e. reflect to themselves.

Property 4. Points not on the line l are reflected to the opposite side of l.

We can take these properties directly as the requirements for our definition of reflecting over a circle, just by replacing line l with circle ω. We have commented on how the naïve definition fails to satisfy **Properties 1, 2,** and **4**, but **Property 3** is satisfied.

Our note about the reflection of segment AB in **Figure** 3, however, warrants another property to be demanded from the upcoming definition:

Property 5. If a point moves along a circle or a line, then its reflection also traces a circle or a line.

We shall introduce *inversion*, which is a possible definition of reflection across a circle satisfying all these properties. In fact, the words are synonymous by convention, so the author apologises for his misuse of terminology.

2.2 Definition of Inversion

The author finds no better way to formulate of the definition of inversion than simply taking an extended quotation from his favorite mathematics textbook, *Euclidean Geometry in Mathematical Olympiads* (2016).

The idea is to view every line as a circle with infinite radius. We add a special point P_∞ to the plane, which every ordinary line passes through (and no circle passes through). This is called the **point at infinity**. Therefore, every choice of three distinct points determine a unique circle or line—three noncollinear ordinary points determine a circle, while two ordinary points plus the point at infinity determine a line.

With this said, we can now define an inversion. Let ω be a circle with centre O and radius r. We say an **inversion** about ω is a map (that is, a transformation) which does the following.

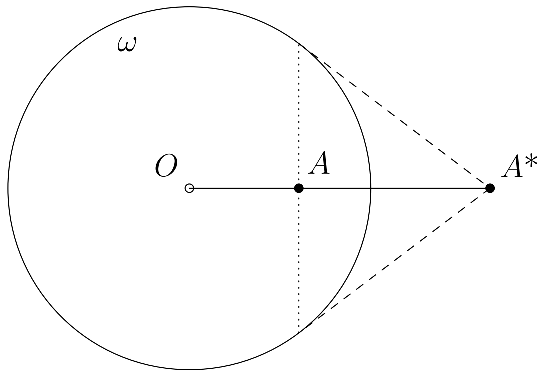

Figure 4: A^* is the image of point A when we perform an inversion about ω

- The centre O of the circle is sent to P_∞
- The point P_∞ is sent to O.
- For any other point A, we send A to the point A^* lying on ray OA such that $OA \cdot OA^* = r^2$.

Quoted at length from [1], with corrections

The definition of P_∞ above facilitates taking an inverse of point O, allowing us to treat lines and circles as highly similar objects. **Figure** 4 shows a geometrical interpretation of the relationship between a point A and its inverse A^*.

A^* is the point where the ray OA intersects the tangent to ω at B, where B is a point on ω such that $\angle BAO = 90°$. Conversely, A is the point where the chord given by the tangents of A^* to ω intersects line OA^*.

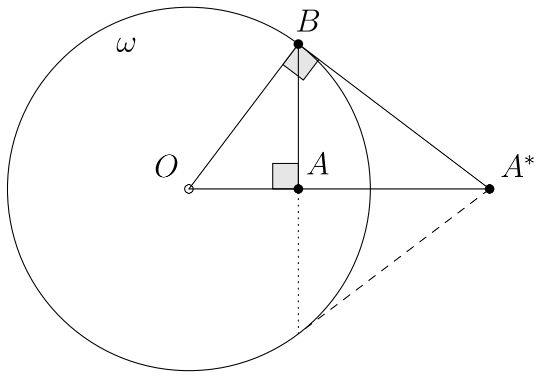

Figure 4.1: A^* is the image of point A when we take an inversion about ω

We can prove this by adding the point of tangency B, and considering triangles $\triangle OAB$ and $\triangle OBA^*$. These triangles are similar, because they both have a right angle and the angle $\angle AOB$. Hence,

$$\frac{OA}{OB} = \frac{OB}{OA^*} \implies OA \cdot OA^* = OB^2 = r^2.$$

2.3 Properties of Inversion

Let us verify that inversion satisfies all the properties demanded in the previous section.

Consider the inversion with respect to the circle ω with centre O and radius r. For any point A, denote its inverse by A^*.

Property 1. The *inverse* of each point exists and is uniquely defined.

Proof. If $A = O$, we explicitly define $A^* = P_\infty$.

Otherwise, A^* is the (unique) point on ray OA whose distance to O is $\frac{r^2}{OA}$, by definition.

Property 2. *Inverting* the *inverse* of a point yields the point itself.

Proof. If $A = O$, then the inverse of its inverse is $(O^*)^* = P_\infty^* = O$.

Otherwise we have equations

$$\begin{cases} OA \cdot OA^* = r^2 \\ OA^* \cdot O\left(A^*\right)^* = r^2 \end{cases} \tag{2.1}$$

which combine to give $OA = O\left(A^*\right)^*$. Then, since A^* and $\left(A^*\right)^*$ are both on ray OA, we necessarily have $A = \left(A^*\right)^*$.

Property 3. Points on the *circle* ω are fixed, i.e. *invert* to themselves.

Proof. Since point A lies on the circle ω, we have $OA = r$, so $OA^* = \frac{r^2}{r} = r$ and so $A = A^*$.

Property 4. Points not on the *circle* ω are *inverted* to the opposite side of ω.

Proof. If point A lies inside the circle, then $OA < r$, so $OA^* = \frac{r^2}{OA} > \frac{r^2}{r} = r$, so its inverse is outside.

Likewise, if point A lies outside, we have $OA > r \implies OA^* < r$, so its inverse is inside, as desired.

We will need more *complex* techniques to execute the proof of **Property 5.**.

2.4 Inversion on the Complex Plane

We introduce a lemma which allows us to invert points on the complex plane over the unit circle, allowing us to algebraically prove many claims about inversion.

Lemma 1. *For point $a \neq 0$ on the complex plane, its inverse with respect to the unit circle is given by* $(\overline{a})^{-1}$

Proof. We show this by observing the modulus and the argument of point $(\overline{a})^{-1}$.

For the modulus, we have

$$\left|(\overline{a})^{-1}\right| = |\overline{a}|^{-1} = |a|^{-1}$$

which implies that $|(\overline{a})^{-1}| \cdot |a| = 1 = 1^2$, and indeed 1 is the radius of the unit circle, so the condition $OA \cdot OA^* = r^2$ is fulfilled.

For the argument,

$$\arg\left((\overline{a})^{-1}\right) = -1 \cdot \arg(\overline{a}) = \arg(a)$$

meaning that indeed $(\overline{a})^{-1}$ lies on the ray from the origin which contains a. Since the origin is the centre of the unit circle, the condition that A^* lies on ray OA is fulfilled, completing the proof. □

Next, we must recall how to characterize lines and circles on the complex plane.

Lemma 2. *A line in $\mathbb{C}$ is the set of all points $z \in \mathbb{C}$ satisfying*

$$\operatorname{Re}(\alpha z) = c \tag{2.2}$$

for some fixed $\alpha \in \mathbb{C} \setminus \{0\}$ and $c \in \mathbb{R}$. Moreover, this line passes through the origin if and only if $c = 0$.

Proof. Write $z = x + iy$ and $\alpha = u + iv$ with $x, y, u, v \in \mathbb{R}$. Then

$$\operatorname{Re}(\alpha z) = \operatorname{Re}((u + iv)(x + iy)) = ux - vy.$$

Hence (2.2) is the linear equation

$$ux - vy = c,$$

which defines a line provided $(u, v) \neq (0, 0)$, i.e. $\alpha \neq 0$. Conversely, any line in the plane has an equation $ux - vy = c$ with $u, v, c \in \mathbb{R}$ not both u, v zero; setting $\alpha = u + iv$ yields (2.2).

Finally, the line passes through the origin if and only if 0 satisfies (2.2), i.e. $\mathrm{Re}(\alpha \cdot 0) = 0 = c$.

Lemma 3. *A circle of radius $r \in \mathbb{R}^+$ and centre $p \in \mathbb{C}$ is the set of all points $z \in \mathbb{C}$ satisfying*

$$|z - p| = r. \tag{2.3}$$

Moreover, the circle passes through the origin if and only if $|p| = r$.

Proof. Write $z = x + iy$ and $p = a + ib$ with $a, b \in \mathbb{R}$. Then

$$|z - p| = |(x - a) + i(y - b)| = \sqrt{(x-a)^2 + (y-b)^2}.$$

Therefore (2.3) is equivalent to

$$(x - a)^2 + (y - b)^2 = r^2,$$

which is the standard Cartesian equation of the circle of radius r centred at (a, b).

Finally, the circle passes through the origin if and only if $z = 0$ satisfies (2.3), i.e. $|0 - p| = |p| = r$.

Now we may proceed to show **Property 5.**, which states that the inverse of a circle or line is a circle or a line. Specifically, we claim the following properties:

Property 5.6. A line passing through O inverts to itself.

Property 5.7. The inverse of a circle not passing through O is a circle not passing through O.

Property 5.8. The inverse of a line not passing through O is a circle passing through O. Equivalently, the inverse of a circle passing through O is a line not passing through O.

Proof. Let ω be the unit circle, so that inversion is given by

$$z \longmapsto (\overline{z})^{-1},$$

and O is the origin. The choice of real and imaginary axes is arbitrary.

Property 5.6. A line passing through O inverts to itself.

Let l be a line through O. By lemma 2, there exists a fixed complex number $\alpha \in \mathbb{C} \setminus \{0\}$ such that each point $z \in l$ is described by the linear equation

$$\mathrm{Re}(\alpha z) = 0.$$

For $z \neq 0$, set $w = (\overline{z})^{-1}$, the inverse of z in the unit circle. Then

$$\mathrm{Re}(\alpha w) = \mathrm{Re}\big(\alpha(\overline{z})^{-1}\big) = \mathrm{Re}\left(\alpha \, \frac{z}{|z|^2}\right) = \frac{1}{|z|^2}\,\mathrm{Re}(\alpha z) = 0,$$

so $w \in l$. Hence the inverse of every point of $l \setminus \{0\}$ lies on l, and since inversion swaps 0 with P_∞ (by definition) which both lie on the line, the whole line l inverts to itself. □

Property 5.7. The inverse of a circle not passing through O is a circle not passing through O.

Let the original circle have centre $p \in \mathbb{C}$ and radius $r_1 > 0$, with $|p| \neq r_1$. It suffices to show that there exist $p' \in \mathbb{C}$ and $r_1' > 0$ such that any point z satisfying

$$|z - p| = r_1$$

also satisfies

$$\big|(\overline{z})^{-1} - p'\big| = r_1'.$$

Recall that for any $w \in \mathbb{C}$,

$$|w|^2 = w\overline{w}. \tag{2.4}$$

Squaring the equation $|z-p| = r_1$ and expanding gives

$$r_1^2 = (z-p)(\overline{z}-\overline{p}) \tag{2.5}$$

$$= z\overline{z} - z\overline{p} - \overline{z}p + |p|^2. \tag{2.6}$$

Rearranging,

$$z\overline{p} + \overline{z}p - z\overline{z} = |p|^2 - r_1^2. \tag{2.7}$$

Since $|p|^2 - r_1^2 \neq 0$, we may multiply both sides of eq. (2.7) by $\dfrac{z^{-1}(\overline{z})^{-1}}{|p|^2-r_1^2}$, obtaining

$$\frac{(\overline{z})^{-1}\overline{p}}{|p|^2-r_1^2} + \frac{z^{-1}p}{|p|^2-r_1^2} - \frac{1}{|p|^2-r_1^2} = z^{-1}(\overline{z})^{-1}. \tag{2.8}$$

Rearranging and factoring,

$$\left(z^{-1} - \frac{\overline{p}}{|p|^2-r_1^2}\right)\left((\overline{z})^{-1} - \frac{p}{|p|^2-r_1^2}\right) = \frac{r_1^2}{(|p|^2-r_1^2)^2}. \tag{2.9}$$

Applying eq. (2.4) once more yields

$$\left|(\overline{z})^{-1} - \frac{p}{|p|^2-r_1^2}\right|^2 = \left(\frac{r_1}{|p|^2-r_1^2}\right)^2.$$

Thus the image is a circle with

$$p' = \frac{p}{|p|^2-r_1^2}, \qquad r_1' = \frac{r_1}{|p|^2-r_1^2}.$$

Moreover, $|p| \neq r_1$ implies $|p'| \neq r_1'$, so the image circle does not pass through O. □

Property 5.8. Lines not through O invert to circles through O, and vice versa.

We first show that a circle passing through O inverts to a line. Let $|z-p| = r_1$ with $|p| = r_1$. Repeating the above argument up to eq. (2.7), we now have

$$z\overline{p} + \overline{z}p - z\overline{z} = 0.$$

Multiplying by $z^{-1}(\overline{z})^{-1}$ gives

$$(\overline{z})^{-1}\overline{p} + z^{-1}p - 1 = 0.$$

Taking real parts,

$$\mathrm{Re}\big(p\, z^{-1}\big) = \tfrac{1}{2}.$$

By lemma 2, this is the equation of a line not passing through O.

Conversely, let a line not passing through O be given by

$$\mathrm{Re}(\alpha z) = c, \qquad c \neq 0.$$

Substituting $z = (\overline{w})^{-1}$ yields

$$\mathrm{Re}\big(\alpha(\overline{w})^{-1}\big) = c,$$

which may be rewritten as

$$\left| w - \frac{\overline{\alpha}}{2c} \right| = \frac{|\alpha|}{2c},$$

the equation of a circle passing through O. This completes the proof. $\square$

Now, let us return to *reality* and marvel at our creation.

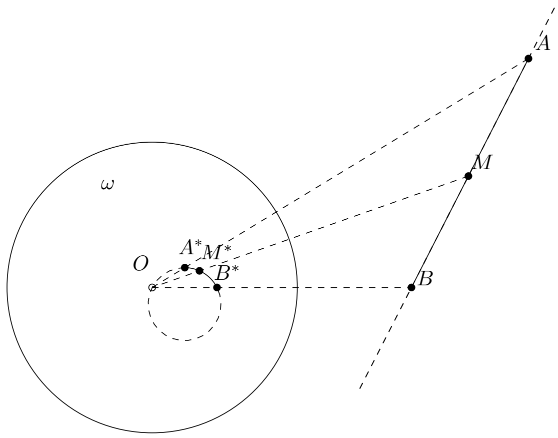

Figure 5: What **Figure** 3 would look like using inversion.

Property 5.8. is demonstrated in **Figure** 5. The inverse of line AB is the circle passing through O, A^*, and B^*, and conversely the inverse of circle (OA^*B^*) is line AB.

It is indeed **Property 5.** that makes inversion an useful tool, able to transform circles into lines, and vice versa.

As a supplement, let us formulate and derive an additional property not mentioned in Chen's masterpiece, wnich is especially helpful in combinatorial tasks.

Lemma 4. *If points A, B are on different sides of circle or line Ξ, then their inverses will be on different sides of the inverse of Ξ.*

We can define the notion of two points A, B lying on different sides of an object Ξ formally, to mean that any path from A to B on the plane must intersect Ξ.

By this definition, A and A' of **Figure** 3 are clearly on the same side of ω, since we can draw a curve connecting them which never touches ω, so that definition would not satisfy this property.

Proof. Assume, for contradiction, that A^* and B^* lie on the same side of the inverse of Ξ. Then there exists a path from A^* to B^* which does not intersect the inverse of Ξ.

Taking inverses pointwise along this path yields a path from A to B which does not intersect Ξ, by **Property 1.**. This contradicts the assumption that A and B lie on different sides of Ξ.

3 Conclusions

In this exploration, the naïve 'equal-distance along the radius' idea for reflecting across a circle was tested against basic expectations of a reflection: being well-defined, involutive, swapping inside/outside, and most importantly keeping simple things simple. It fails most sharply on the last point, producing complicated images from straight or circular motion. Inversion replaces that definition with a constrained rule (O, A, A^* collinear and $OA \cdot OA^* = r^2$, plus a point at infinity), and the resulting map behaves exactly as expected: it is involutive, fixes the reference circle, and preserves circle and lines as circles and lines, as shown algebraically on the complex plane. This matches the aim: to justify inversion as a natural 'reflection across a circle'.

References

[1] Evan Chen. *Euclidean Geometry in Mathematical Olympiads*. MAA Problem Books, 311 pp. Washington, DC: Mathematical Association of America, 2016. ISBN: 9780883858394.